Review. 03/13 F ①

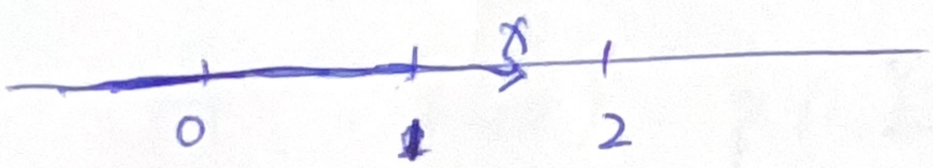
pdf $\rightarrow$ cdf $\qquad F(x) = \int_{-\infty}^{x} f(x)\,dx$

$f(x)$ $\qquad F(x)$

$$f(x) = \begin{cases} x & 0 \le x < 1 \\ 2-x & 1 \le x < 2 \\ 0 & \text{otherwise.} \end{cases}$$

for $0 \le x < 1$ $\qquad F(x) = \int_{-\infty}^{x} f(x)\,dx = \int_{0}^{x} f(x)\,dx$

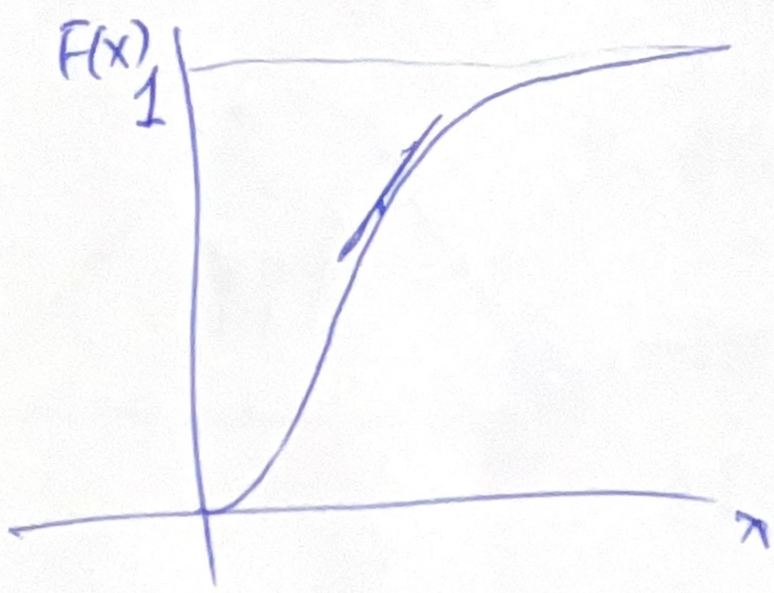
for $1 \le x < 2$ $\qquad F(x) = \int_{-\infty}^{x} f(x)\,dx = \boxed{\int_{0}^{1} f(x)\,dx} + \int_{1}^{x} f(x)\,dx$



for $x < 0$ $\qquad F(x) = 0$

for $x \geq 2$. $\qquad F(x) = 1.$

cdf $F(x) \rightarrow$ pdf $f(x)$ $\qquad \boxed{f(x) = \boxed{\frac{d}{dx}} F(x)}$



$$E(X) = \int_{-\infty}^{\infty} x\, f(x)\,dx$$

$$Var(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

R.S $\Rightarrow$ iid. independent and identically distributed.

a) $Y = X_1 + X_2 + \cdots + X_{25}$.

Find distribution of Y.

$$Y \sim N\Big(\mu_Y = \underbrace{160 + 160 + \cdots + 160}_{25} = 160 \times 25 = 4000,$$

$$\sigma_Y^2 = \underbrace{30^2 + 30^2 + \cdots + 30^2}_{25} = 30^2 \times 25 = 150^2\Big)$$

b) Elevator has a weight limit of 4300 pounds.

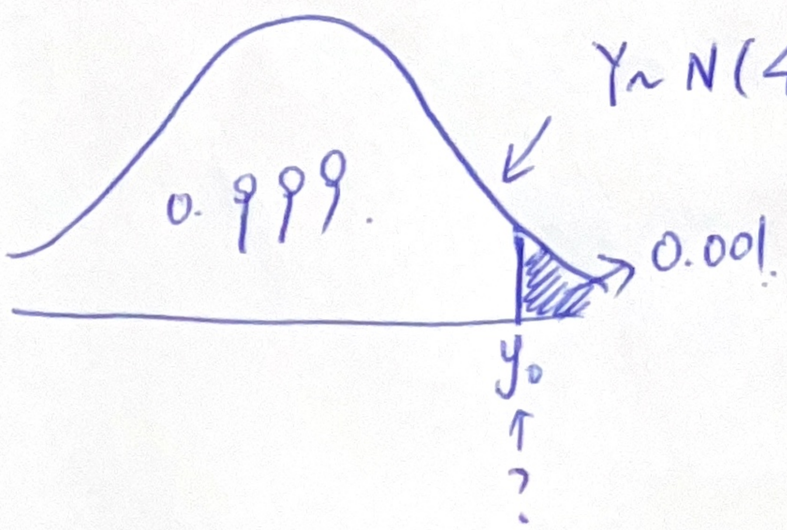
Find probability that 25 R.S. people will exceed limit.

$$P(Y > 4300) = P\left(\boxed{\frac{Y-4000}{150}} > \frac{4300-4000}{150}\right)$$

$$= P(Z > 2)$$

$$= 1 - P(Z \le 2) = 0.0228.$$

c) What should weight limit be such that probability of 25 R.S people exceed that weight is less than 0.001



$$\underline{0.999} = P(Y \le y_0)$$

$$= P\left(\boxed{\frac{Y-4000}{150}} \le \frac{y_0 - 4000}{150}\right)$$

$$= P\left(Z \le \frac{y_0 - 4000}{150}\right)$$